

STOREAM
(Store & Stream OTT platform)

SOFTWARE PROJECT REPORT
submitted for the partial fulfilment of the award of the degree of
MASTER OF COMPUTER APPLICATIONS

BY

ASWIN A
BP231033

Under the Guidance of
Prof. Dr. S. Anthony Philomen Raj M.C.A, Ph. D.



PG DEPARTMENT OF COMPUTER APPLICATION
SACRED HEART COLLEGE (AUTONOMOUS)
TIRUPATTUR-635 601

APRIL 2025

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This is to certify that the project entitled " **STOREAM (Store & Stream)** " being submitted to Sacred Heart College (Autonomous), Tirupattur by **ASWIN A (BP231033)** for the partial fulfilment for the award of a degree in Master of Computer Applications is a project work carried out under my guidance and supervision.

Signature of the HOD

Signature of the Guide

Submitted for the viva-voce examination held on: _____

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1:

2:



Department of Computer Applications (PG)

SACRED HEART COLLEGE (Autonomous)

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- III. The project has been checked using TURNITIN (copy of originality report attached as an Annexure at the end of the project) and found within limits (less than 30%) as per Sacred Heart College Plagiarism policy

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Project Guide

Dr. A. George Louis Raja

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ACKNOWLEDGEMENT

I thank God Almighty for the blessings and grace by which I was able to complete this project work successfully. I sincerely thank my parents, unless and my relatives, for their continuous support, love and blessings for me to attain many achievements.

I take this opportunity to express my gratitude to those who helped me and led me to complete this project work. I offer my humble gratitude to **Rev. Dr. D. Maria Antony Raj Principal**, Sacred Heart College (Autonomous), Tirupattur for permitting me to do my project work. I would like to thank the Director, HR and the entire team at Queenbug Technologies for the guidance they provided to me to complete this project successfully.

I express my deep gratitude to **Dr. S. Sagayaraj** Director, Department of Computer Applications (PG), Sacred Heart College (Autonomous), Tirupattur.

I would like to thank **Dr. A. George Louis Raja** Head, Department of Computer Applications (PG), Sacred Heart College (Autonomous) Tirupattur for providing the necessary arrangements for me to do my project work and training at Queenbug Technologies

My sincere thanks to **Asst. Prof. Anthony Philomen Raj** internship Guide, who has guided me throughout my project work, I am grateful to him for his encouragement, guidance and assistance towards my project work.

ASWIN A
(BP231033)

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CHAPTER I

Abstract:

Stoream is an innovative **Over-The-Top (OTT) Platform** designed to revolutionize the way users stream and manage movies based on their subscription plans. The platform caters to a diverse audience by offering **three distinct subscription tiers: Basic, Pro, and Enterprise**. Each tier provides users with varying levels of access to **movie categories** and **storage limits** for uploading their own content.

- **Basic Plan:** Users gain access to **Tamil Movies** and are allocated **1GB of free storage** for uploading their personal movies.
- **Pro Plan:** Users enjoy access to **Tamil Movies** and **Web Series**, along with **2GB of free storage** for uploads.
- **Enterprise Plan:** Users have access to all categories, including **Tamil Movies, Web Series, Animation, and Hollywood Movies**, and are provided with **5GB of free storage** for uploads.

Stoream ensures **efficient storage management** by enforcing strict **storage limits** and **file size restrictions**. Ensuring fair usage of resources and preventing server overload. This feature is particularly important for maintaining platform performance and scalability.

The platform also includes a robust **admin panel** that allows administrators to perform **CRUD (Create, Read, Update, Delete) operations** on **movie categories** and **user accounts**. However, to ensure **user privacy** and **data security**, admins are **restricted from accessing user-uploaded movies**. This design choice ensures that user content remains private and secure.

Built with a **React-based frontend**, Stoream delivers a **dynamic and responsive user interface** that enhances the user experience. The frontend communicates seamlessly with the **PHP backend**, which handles **file uploads, storage management, and database operations**. The backend is designed to enforce storage limits, manage user subscriptions, and ensure smooth file handling.

Introduction:

Stoream is an innovative **Over-The-Top (OTT) Platform** designed to provide users with a seamless and personalized experience for streaming and managing movies based on their subscription plans. In today's digital age, where content consumption is at an all-time high, Stoream aims to cater to the diverse needs of users by offering a flexible and feature-rich platform. Whether you are a casual viewer or a movie enthusiast, Stoream ensures that you have access to the content you love, while also providing tools to manage your own uploaded movies.

The platform is built around the concept of **subscription-based access**, offering **three distinct plans: Basic, Pro, and Enterprise**. Each plan is tailored to meet the needs of different user segments, providing varying levels of access to **movie categories** and **storage limits** for personal uploads. Stoream not only focuses on delivering a rich streaming experience but also emphasizes **efficient storage management** and **user privacy**.

Key Features of Stoream

1. Subscription Plans:

- **Basic Plan:** Access to **Tamil Movies** and **1GB of free storage** for personal uploads.
- **Pro Plan:** Access to **Tamil Movies** and **Web Series**, along with **2GB of free storage**.
- **Enterprise Plan:** Access to all categories, including **Tamil Movies, Web Series, Animation, and Hollywood Movies**, with **5GB of free storage**.

2. Storage Management:

- Enforces **storage limits** based on the user's subscription plan.

3. Admin Panel:

- Provides administrators with tools to manage **movie categories** and **user accounts**.
- Ensures **user privacy** by restricting admin access to user-uploaded movies.

4. User Experience:

- A **React-based frontend** delivers a **dynamic and responsive user interface**.
- Users can easily **stream movies, upload personal content, and manage their storage**.
- Real-time feedback on storage usage and available space.

5. Security and Privacy:

- Admins can perform **CRUD operations** on movie categories and user accounts but **cannot access user-uploaded movies**.
- Ensures that user content remains private and secure.

1.1 Purpose

The purpose of the **Stoream OTT Platform** is to:

- Provide users with a **streaming platform** tailored to their subscription plans.
- Allow users to **upload and manage their own movies** within their allocated storage limits.
- Offer **different movie categories** based on the user's subscription tier:
 - **Basic Plan:** Tamil Movies.
 - **Pro Plan:** Tamil Movies and Web Series.
 - **Enterprise Plan:** Tamil Movies, Web Series, Animation, and Hollywood Movies.
- Provide an **admin panel** for managing movie categories and user accounts while ensuring **privacy** for user-uploaded content.
- Enforce **storage limits** and **file size restrictions** to ensure fair resource usage.

1.2 System Overview and Development Goal

System Overview

The **Stoream OTT Platform** consists of:

1. Frontend:

- Built with **React** for a dynamic and responsive user interface.
- Uses **Bootstrap** for styling and layout.
- Communicates with the backend via **Axios** for API requests.

2. Backend:

- Built with **PHP** for handling file uploads, storage management, and database operations.
- Uses **MySQL** to store user details, movie metadata, and storage information.
- Enforces **storage limits** and **file size restrictions**.

3. Database:

- Stores user information, movie metadata, and storage usage.
- Tracks user plans (Basic, Pro, Enterprise) and their respective storage limits.

4. Admin Panel:

- Allows admins to perform **CRUD operations** on movie categories and user accounts.
- Ensures that admins **cannot access user-uploaded movies**.

Development Goal

The primary goals of the system are:

- To provide a **user-friendly interface** for streaming and managing movies.
- To enforce **storage limits** and **file size restrictions** to ensure fair resource usage.
- To ensure **scalability** and **security** in handling user data and file uploads.
- To provide **real-time feedback** on storage usage and available space.

- To offer an **admin panel** for managing movie categories and user accounts while maintaining user privacy.

1.3 Problem Statement

The current challenges in managing user-uploaded content and subscription-based OTT platforms include:

1. **Storage Overuse:** Users may exceed their allocated storage limits, leading to resource mismanagement.
2. **File Size Restrictions:** Large file uploads can strain server resources and affect platform performance.
3. **Category Management:** Admins need a streamlined way to manage movie categories and user accounts.
4. **User Privacy:** Admins should not have access to user-uploaded content to ensure privacy and security.

The **Stoream OTT Platform** addresses these challenges by:

- Enforcing **storage limits** and **file size restrictions**.
- Providing an **admin panel** for efficient category and account management.
- Ensuring **privacy** by restricting admin access to user-uploaded movies.

1.4 System Model

The **Stoream OTT Platform** follows a **client-server architecture**:

1. **Client (Frontend):**
 - Users interact with the platform via a **React-based web application**.
 - Admins access the **admin panel** for managing categories and accounts.
2. **Server (Backend):**
 - Handles file uploads, storage management, and database operations using **PHP**.
 - Enforces **storage limits** and **file size restrictions**.
3. **Database:**

- Stores user information, movie metadata, and storage usage in a **MySQL database**.

4. Admin Panel:

- Allows admins to perform **CRUD operations** on movie categories and user accounts.
- Ensures that admins **cannot access user-uploaded movies**.

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3. Database:

- Stores user information, movie metadata, and storage usage in a **MySQL database**.

4. Admin Panel:

- Allows admins to perform **CRUD operations** on movie categories and user accounts.
- Ensures that admins **cannot access user-uploaded movies**.

1.5 Project Timeline Diagram (Gantt Chart)

Below is a **Gantt Chart** outlining the project timeline

S. No	Title	Description	Date
1.	Reporting to the company	Going to the company	22/12/2024
2.	Choosing the Project Title	Confirmation of the title of the project with the guides	23/12/2024
3.	Requirements Document	Completing the User Requirement and System Requirement Documents	28/12/2024
4.	Design Document	Finalizing and Completing the Design Document	29/12/2024
5.	UI Design	Completing the UI Design for the project	31/12/2024
6.	Database Design	Creating and completing the Database Design for the project	02/01/2025
7.	Project Implementation	Implementation of the project and deployment are completed	20/02/2025
8.	Testing	After Implementation, the testing process of the project is completed	24/02/2025
9.	Documentation	Completing the Project Documentation	28/02/2025
10.	First Review	First Review of the Project	07/03/2025

1.6 ORGANIZATION PROFILE

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Queenbug Technologies is a leading provider of a wide range of IT solutions such as Web design and development, e-commerce website development, Mobile Application development, and Digital Marketing services we also incorporate other services like hosting, graphic designing and more to render the right product to the customers. Our expert developers and designers will create the best solution as per the market standards. We have successfully done several projects in website development, Mobile app development, e-commerce website and more by understanding every detail of client's business requirement. We are a Professional mobile application Development company in India, Bangalore, Web Applications and Web Development Company among the Top Development Companies offering affordable mobility solutions.

We are a result-oriented and performance-driven company that always believes in efforts and strives hard to make the impossible possible. We take up every challenge and find the intended results for it. Keeping in mind the tough competition and future technology enhancement, our trained & skilled team of experts will always create the product that blends with state-of-the-art technology. Such that our customers excel among the other competitors and attain the zenith place in the challenging business environment. We believe our success is our customers' growth and always want to maintain a sustaining relationship with them.

OUR SERVICES:

Queenbug Technologies continuously strives to provide high-quality products and services to customers that enable interactive communication between their business and audiences. QUEENBUG TECHNOLOGIES has a typical client base from small to Fortune 70 Companies and offers a one-stop solution for all types of technology services. Our experts specialize in Mobile Application Development Companies in Bangalore, Android OS App Development, Mobile App Development Companies in Bangalore, Top Mobile app development Companies, Mobile app developers, Mobile App Development Company in Bangalore Android Application Development Companies in Bangalore, OS Application Development Companies in Bangalore, Hybrid Mobile App Development Company in Bangalore, Best Mobile App Development Company in Bangalore, Android App Development Company, Mobile Application App Development Company, Mobile Application Development, Web Design and Development, E-commerce Website

Development across various latest technology platforms and always keen on updating with new trends arising in the Industry

1. UI/UX Design

Queenbug specializes in crafting intuitive and engaging user interfaces and experiences that captivate and retain users. Our UI/UX design services focus on usercentric design principles, ensuring seamless navigation, visual appeal, and optimal usability. By leveraging the latest design trends and tools, we help organizations enhance user satisfaction, drive engagement, and achieve their business goals through visually stunning and functionally superior digital products.

2. Mobile Application & Development

We empower businesses to reach their mobile audience with innovative and highperformance mobile applications. Our expert developers create custom mobile solutions tailored to your unique needs, whether for iOS, Android, or cross-platform development. From concept to deployment, we ensure that your mobile app is secure, scalable, and delivers an exceptional user experience, helping you stay competitive in the ever-evolving mobile landscape.

3. Digital Marketing

Queenbug offers comprehensive digital marketing services designed to boost your online presence and drive measurable results. Our strategies encompass search engine optimization (SEO), pay-per-click (PPC) advertising, social media marketing, content marketing, and email campaigns. By leveraging data-driven insights, we help you reach your target audience, increase brand visibility, and achieve higher conversion rates, ultimately maximizing your return on investment.

4. Cloud Migration

Our cloud migration services enable organizations to transition their IT infrastructure and applications to the cloud seamlessly. We provide end-to-end migration solutions, including assessment, planning, execution, and optimization, ensuring minimal disruption to your operations. By migrating to the cloud, businesses can benefit from enhanced scalability, cost efficiency, and improved performance, positioning themselves for future growth and innovation.

5. Embedded Systems

We specialize in the design and development of embedded systems tailored to meet the specific needs of various industries. Our expertise spans hardware design, firmware development, and software integration, ensuring robust and reliable embedded solutions. Whether for IoT devices, industrial automation, or consumer electronics, our embedded systems services deliver high-performance, low-power solutions that drive innovation and efficiency.

6. Web Hosting & Maintenance

Queenbug provides reliable and secure web hosting solutions to ensure your website is always accessible and performing optimally. Our hosting services include scalable infrastructure, robust security measures, and regular backups to safeguard your data. Additionally, our maintenance services encompass ongoing updates, performance monitoring, and technical support, ensuring your website remains up-to-date, secure, and aligned with your evolving business needs.

CHAPTER II

2.1 Module Identification and Description

The **Stoream OTT Platform** is divided into the following key modules:

1. **User Module:**

- **Description:** Handles user registration, login, and profile management.
- **Features:**
 - Users can sign up and choose a subscription plan (Basic, Pro, Enterprise).
 - Users can log in to access their dashboard.
 - Users can view their storage usage and available space.

2. **Subscription Module:**

- **Description:** Manages subscription plans and their associated features.
- **Features:**
 - Users can upgrade or downgrade their subscription plans.
 - Each plan provides access to specific movie categories and storage limits.

3. **Movie Streaming Module:**

- **Description:** Allows users to stream movies based on their subscription plan.
- **Features:**
 - Users can browse and stream movies in their accessible categories.
 - Movies are categorized into Tamil Movies, Web Series, Animation, and Hollywood Movies.

4. **Upload Module:**

- **Description:** Enables users to upload their own movies within their storage limits.
- **Features:**
 - The system enforces storage limits based on the user's subscription plan.

5. **Admin Module:**

- **Description:** Provides administrators with tools to manage the platform.

- **Features:**
 - Admins can perform **CRUD operations** on movie categories and user accounts.
 - Admins **cannot access user-uploaded movies** to ensure privacy.

6. **Storage Management Module:**

- **Description:** Tracks and enforces storage limits for user uploads.
- **Features:**
 - Calculates used and available storage space for each user.
 - Prevents uploads that exceed the user's storage limit.

2.2 User Role and Characteristics

The system has two primary user roles:

1. **Users:**

- **Characteristics:**
 - Can be **Basic, Pro, or Enterprise** subscribers.
 - Can stream movies based on their subscription plan.
 - Can upload personal movies within their storage limits.
 - Can view their storage usage and available space.
- **Responsibilities:**
 - Manage their profile and subscription.
 - Upload and manage their personal movies.

2. **Admin:**

- **Characteristics:**
 - Has full access to manage movie categories and user accounts.
 - Cannot access user-uploaded movies to ensure privacy.
- **Responsibilities:**
 - Perform **CRUD operations** on movie categories.
 - Manage user accounts (create, update, delete).
 - Monitor system performance and resolve issues.

2.3 Functional Requirements

The functional requirements of the **Stoream OTT Platform** are as follows:

1. User Registration and Login:

- Users can register and log in to the platform.
- Users can choose a subscription plan during registration.

2. Subscription Management:

- Users can upgrade or downgrade their subscription plans.
- Each plan provides access to specific movie categories and storage limits.

3. Movie Streaming:

- Users can browse and stream movies based on their subscription plan.
- Movies are categorized into Tamil Movies, Web Series, Animation, and Hollywood Movies.

4. File Upload:

- The system enforces storage limits based on the user's subscription plan.

5. Storage Management:

- The system tracks and displays used and available storage space for each user.
- Prevents uploads that exceed the user's storage limit.

6. Admin Panel:

- Admins can perform **CRUD operations** on movie categories and user accounts.
- Admins cannot access user-uploaded movies.

2.4 Performance Requirements

1. Scalability:

- The system should handle a growing number of users and movies without performance degradation.

2. Response Time:

- The system should respond to user requests within **2 seconds** under normal load.

3. Storage Efficiency:

- The system should efficiently manage storage space and enforce storage limits.

4. File Upload Speed:

- The system should support file uploads with a minimum speed of **5MB/s**.

5. Availability:

- The system should have an uptime of **99.9%**, ensuring it is available for users at all times.

2.5 Hardware and Software Details

Implementation Side

1. Frontend:

- **Framework:** React.js
- **Styling:** Bootstrap
- **API Communication:** Axios

2. Backend:

- **Language:** PHP
- **Database:** MySQL
- **File Storage:** Local server storage for user uploads.

3. Development Tools:

- **IDE:** Visual Studio Code
- **Version Control:** Git and GitHub

Deployment Side

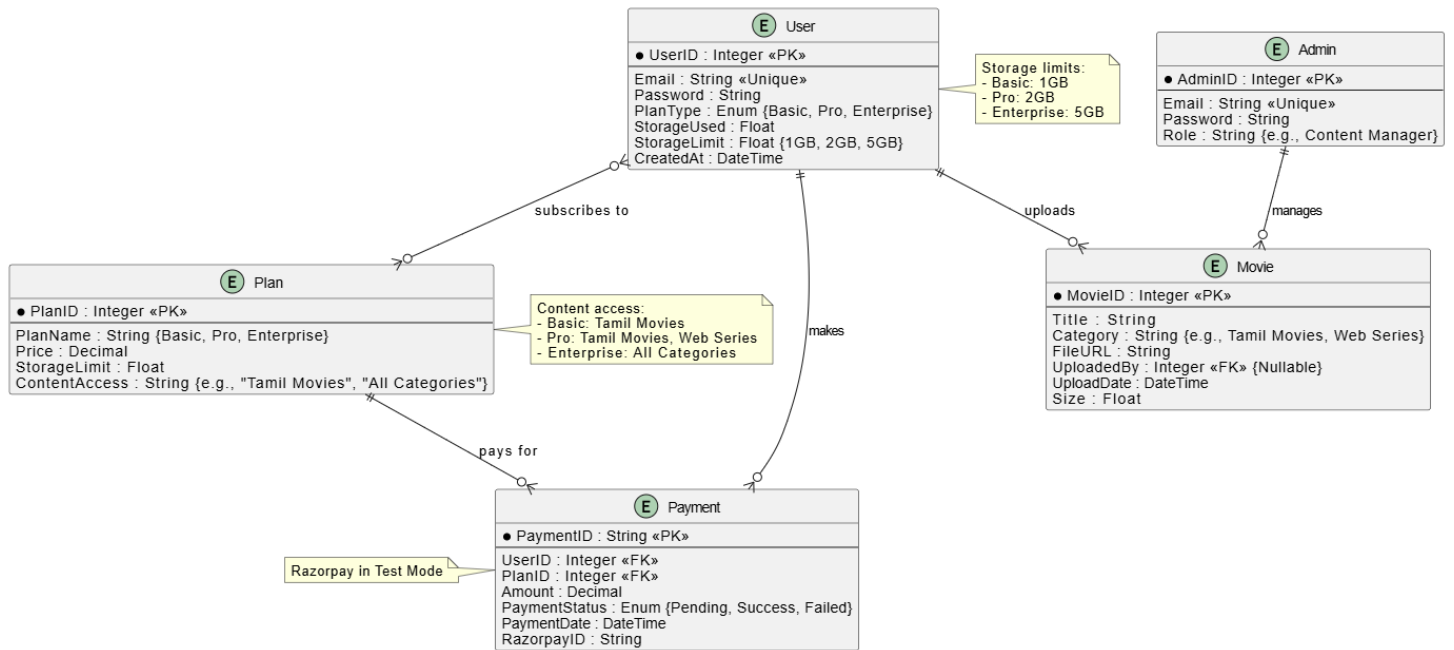
1. Server:

- **Operating System:** Ubuntu Server (Linux)
- **Web Server:** Apache
- **Database Server:** MySQL

2. Hardware Requirements:

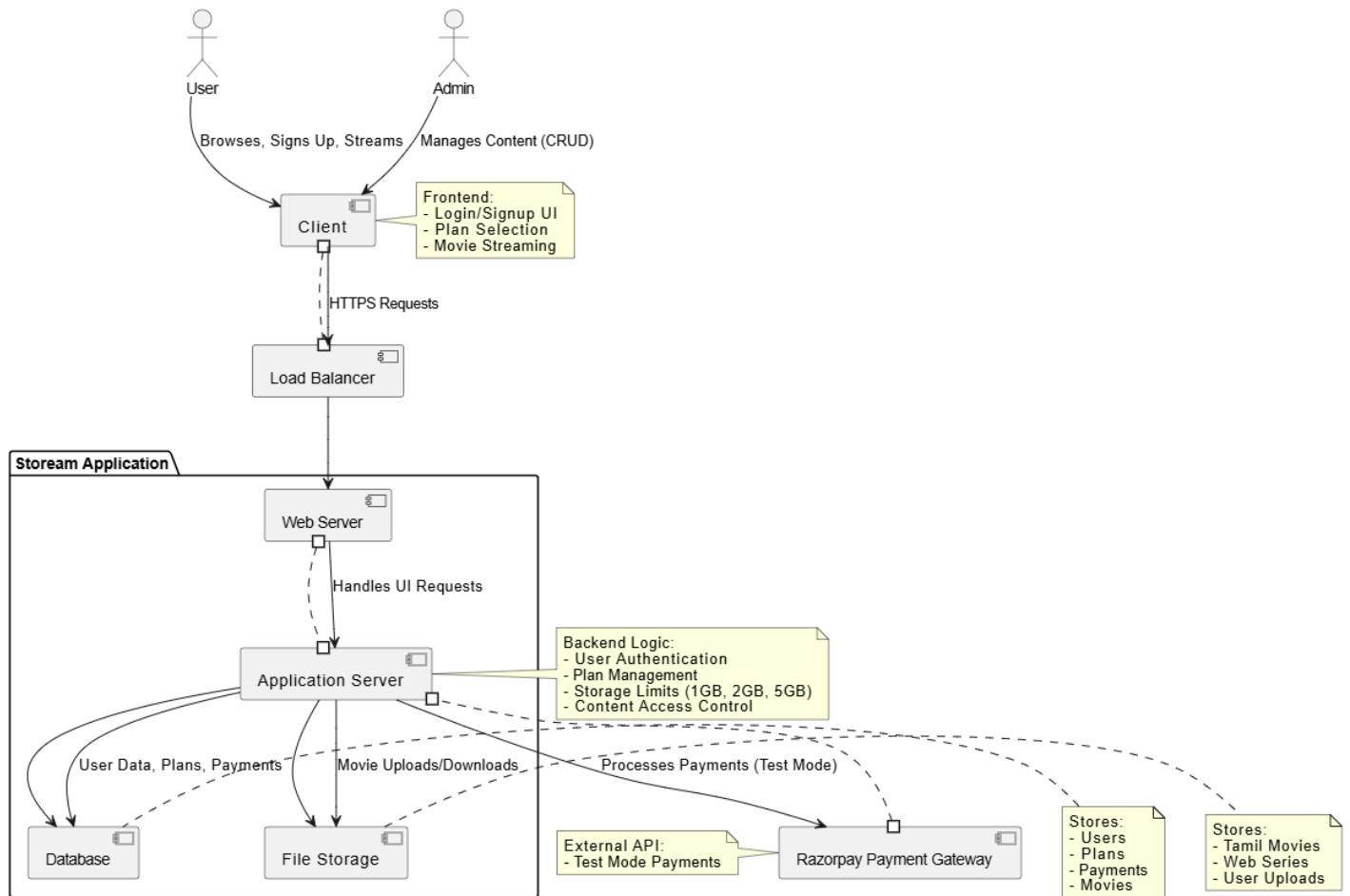
- **Processor:** Intel (8 cores, 2.5 GHz)
- **RAM:** 8GB or higher
- **Storage:** 500GB SSD (for file storage and database)

2.6 Conceptual Diagram (Database ER Diagram)



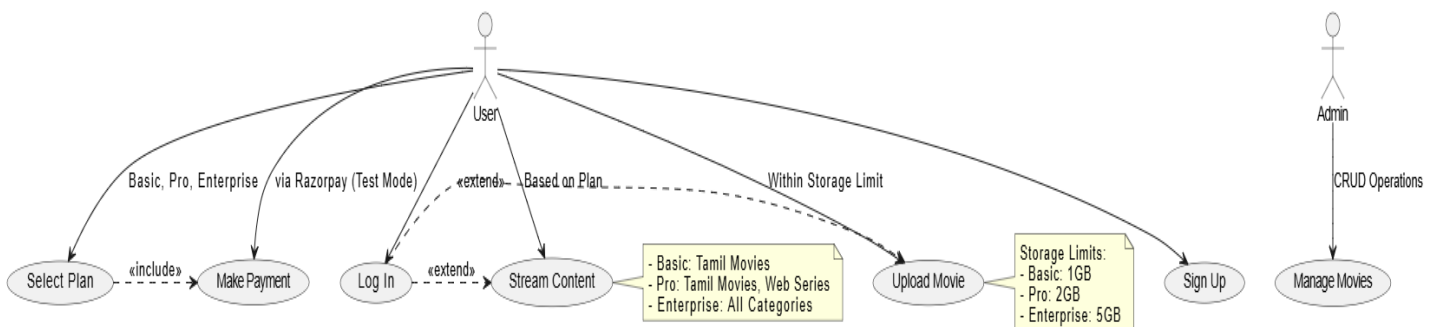
CHAPTER III

3.1 System Architecture Diagram:

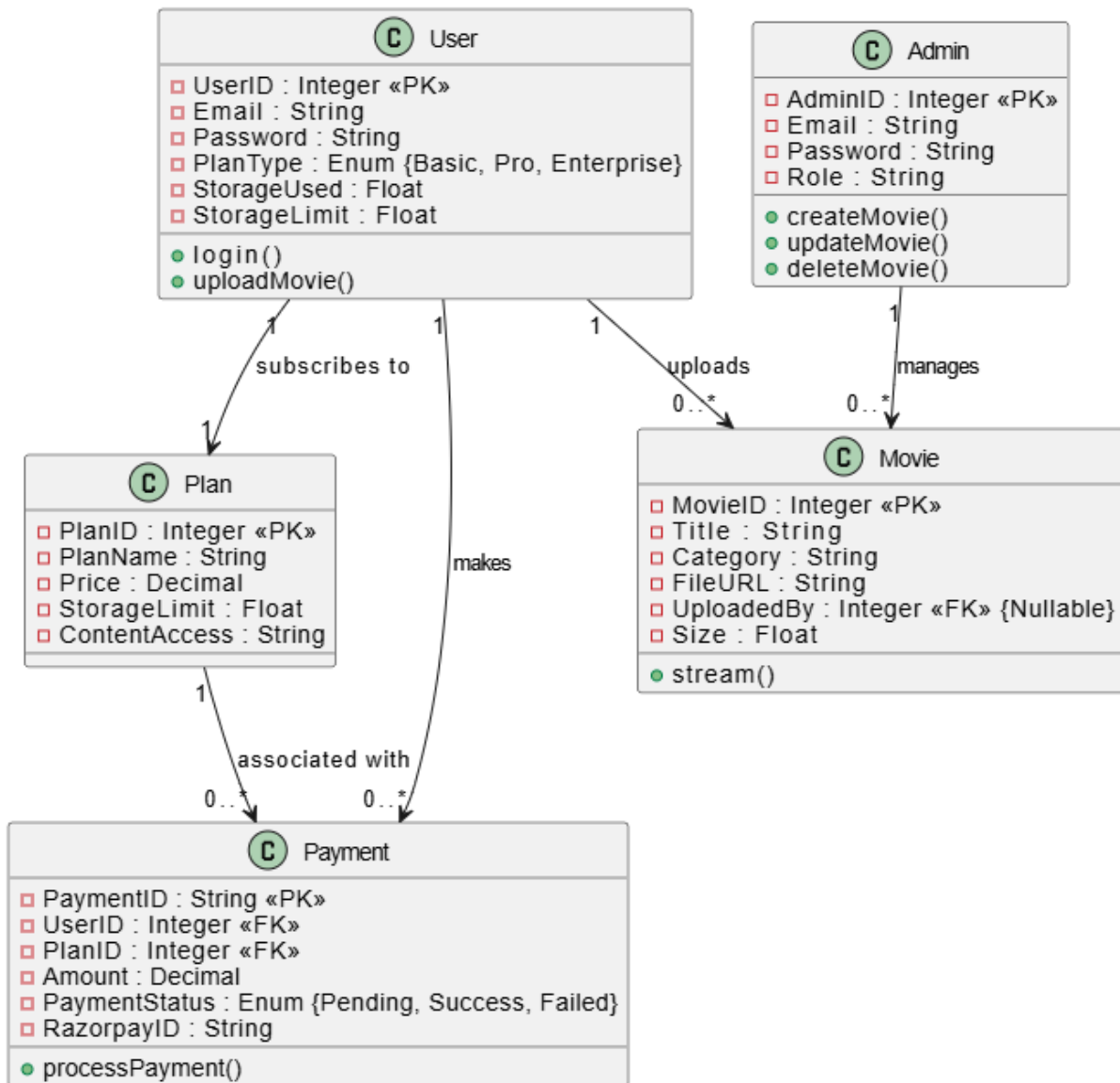


3.2 UML Design Models

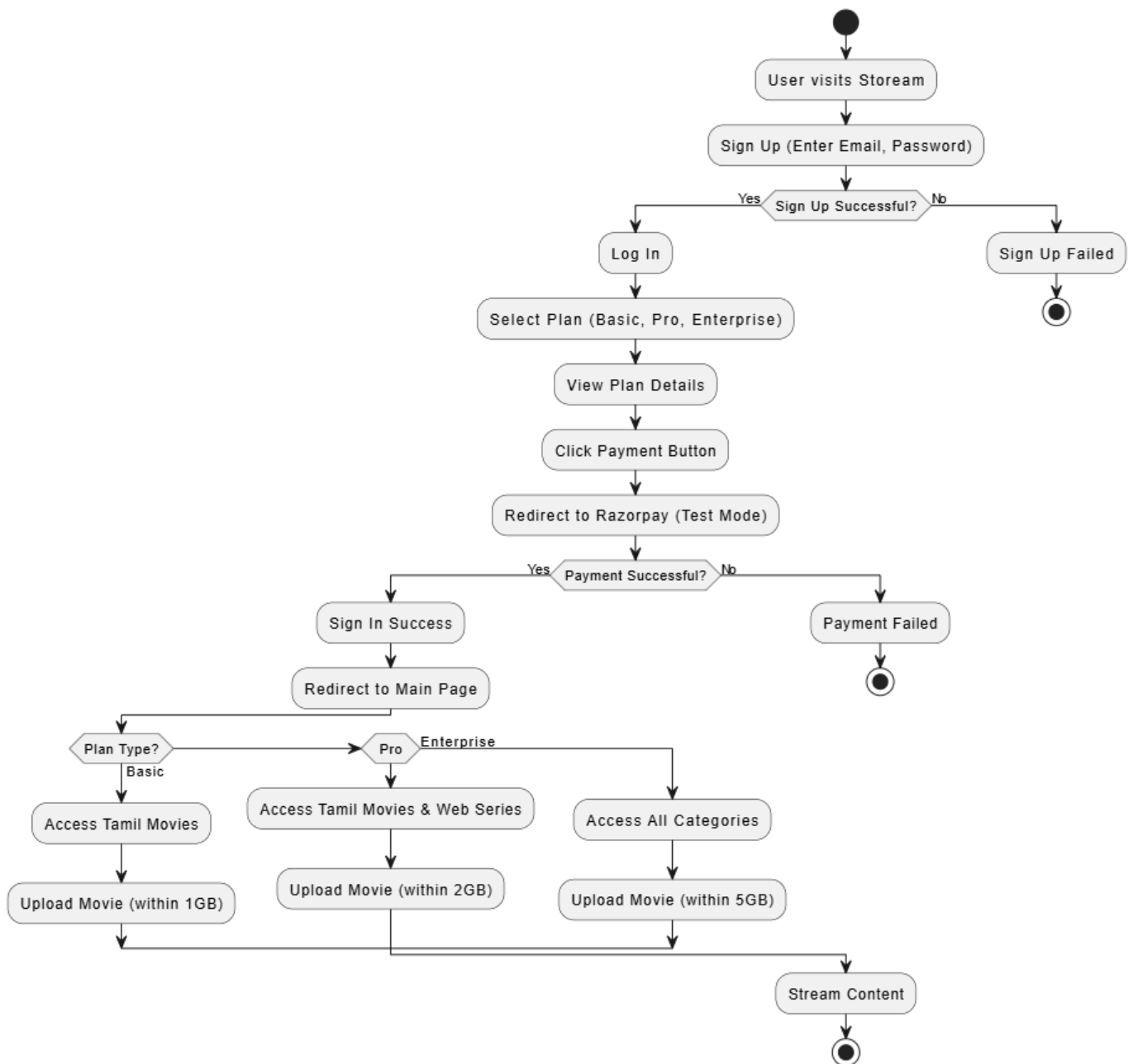
3.2.1 Use Case Diagram



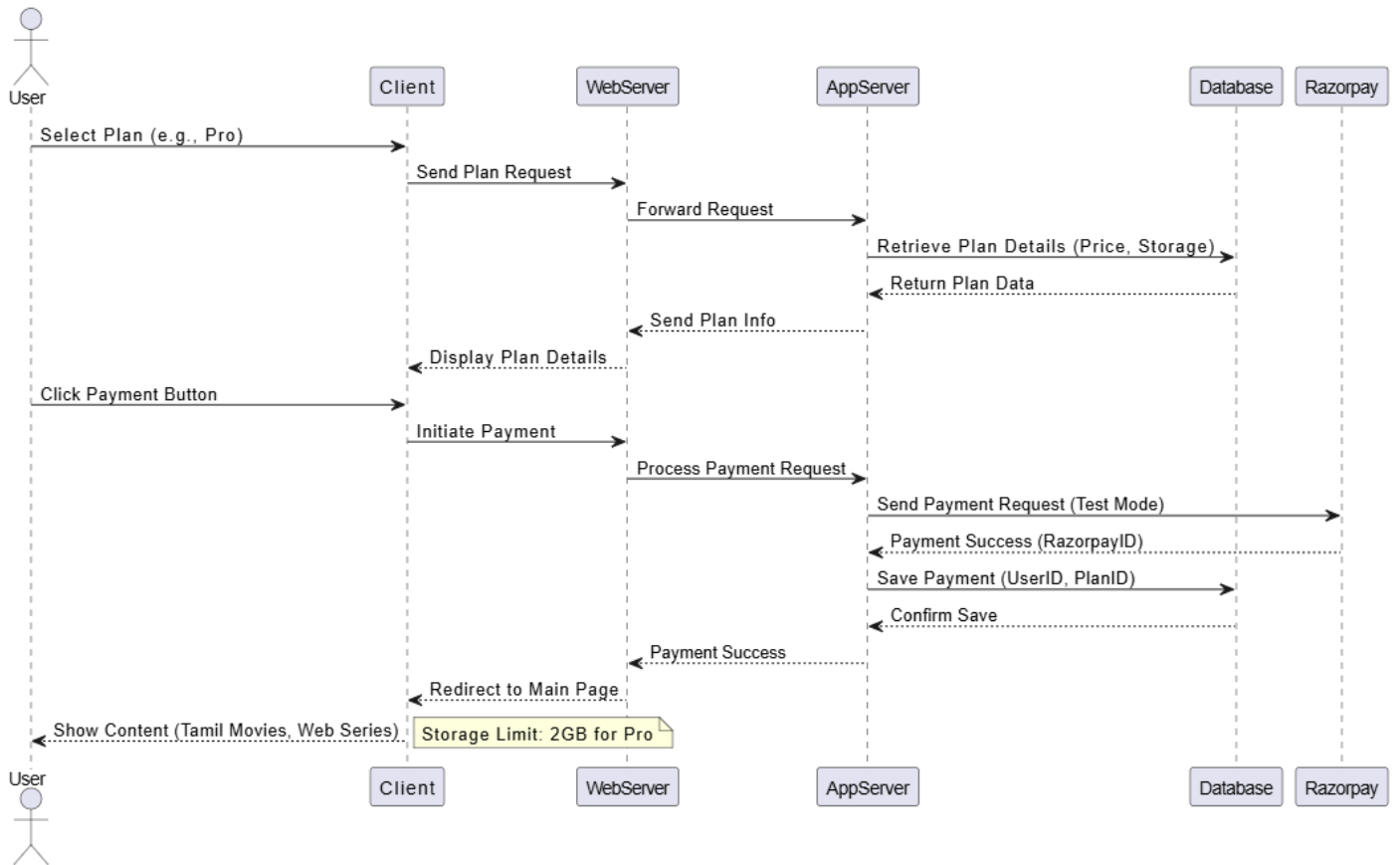
3.2.2 Class Diagram



3.2.3 Activity Diagram



3.2.4 Sequence Diagram



3.3 Database Design Models

3.3.1 Table Structure

1. users_movies Table

Column Name	Data Type	Constraints	Description
id	INTEGER	Primary Key, Auto Increment	Unique identifier for each uploaded movie
user_id	INTEGER	Foreign Key (users.id), Not Null	Links to the user who uploaded the movie
file_name	VARCHAR(255)	Not Null	Name of the uploaded file
file_path	VARCHAR(255)	Not Null	Storage path or URL to the movie file
file_size	FLOAT	Not Null	Size of the file in GB
upload_date	DATETIME	Default: NOW()	Timestamp of movie upload

2. users Table

Column Name	Data Type	Constraints	Description
id	INTEGER	Primary Key, Auto Increment	Unique identifier for each user
first_name	VARCHAR(100)	Not Null	User's first name

last_name	VARCHAR(100)	Not Null	User's last name
email	VARCHAR(255)	Unique, Not Null	User's email for authentication
password	VARCHAR(255)	Not Null	Encrypted password
phone_number	VARCHAR(20)	Nullable	User's contact number (optional)
profile_image	VARCHAR(255)	Nullable	Path or URL to profile image (optional)

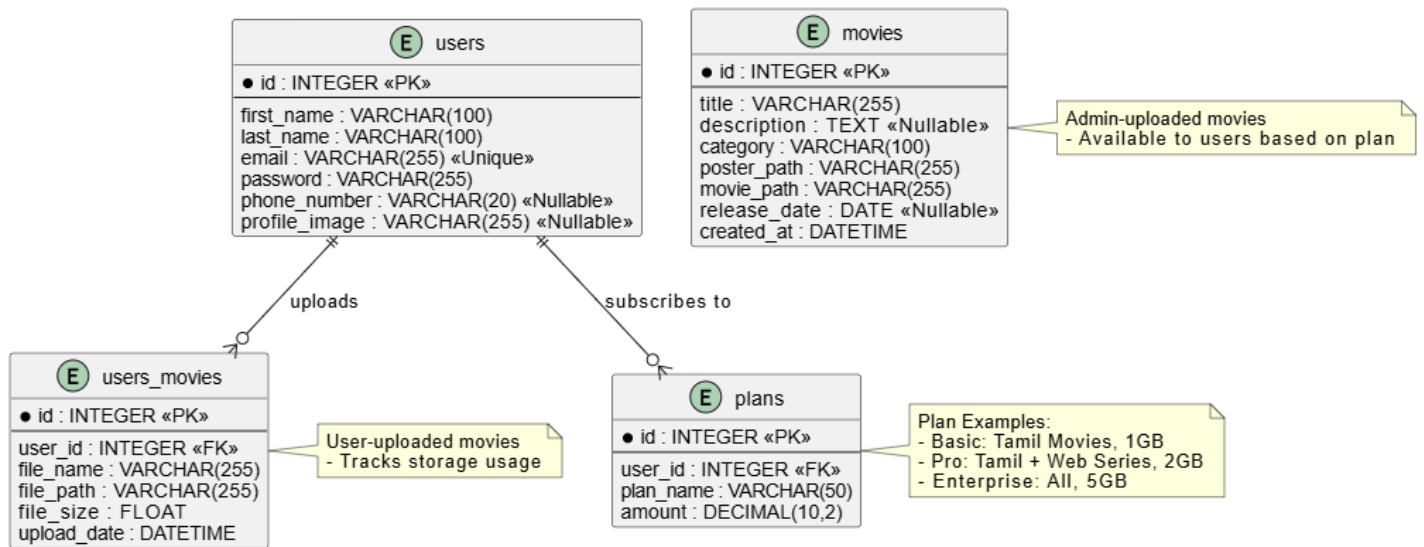
3. plans Table

Column Name	Data Type	Constraints	Description
id	INTEGER	Primary Key, Auto Increment	Unique identifier for each plan record
user_id	INTEGER	Foreign Key (users.id), Not Null	Links to the user subscribed to the plan
plan_name	VARCHAR(50)	Not Null	Name of the plan (e.g., Basic, Pro)
amount	DECIMAL(10,2)	Not Null	Cost of the plan (e.g., 99.99)

4. movies Table (Admin-uploaded movies)

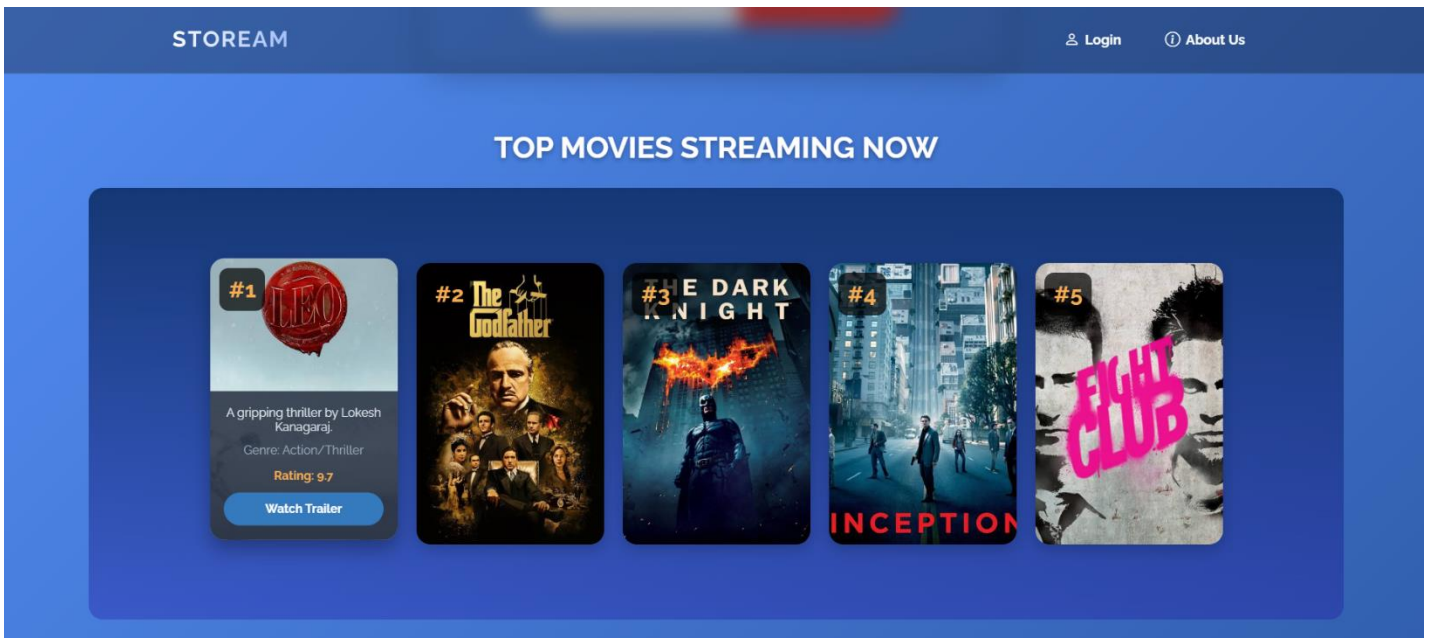
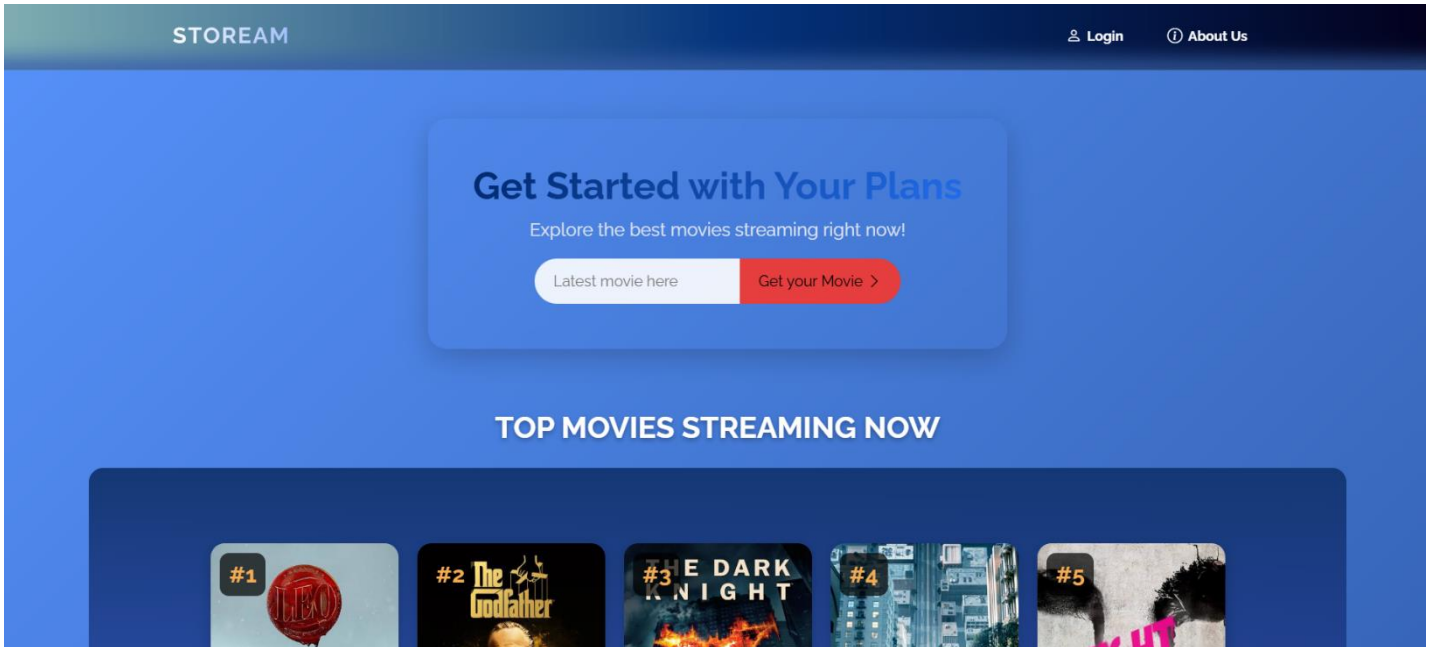
Column Name	Data Type	Constraints	Description
id	INTEGER	Primary Key, Auto Increment	Unique identifier for each movie
title	VARCHAR(255)	Not Null	Movie title
description	TEXT	Nullable	Brief description of the movie
category	VARCHAR(100)	Not Null	Content type (e.g., Tamil Movies)
poster_path	VARCHAR(255)	Not Null	Path or URL to the movie poster
movie_path	VARCHAR(255)	Not Null	Path or URL to the movie file
release_date	DATE	Nullable	Release date of the movie (optional)
created_at	DATETIME	Default: NOW()	Timestamp of movie addition by admin

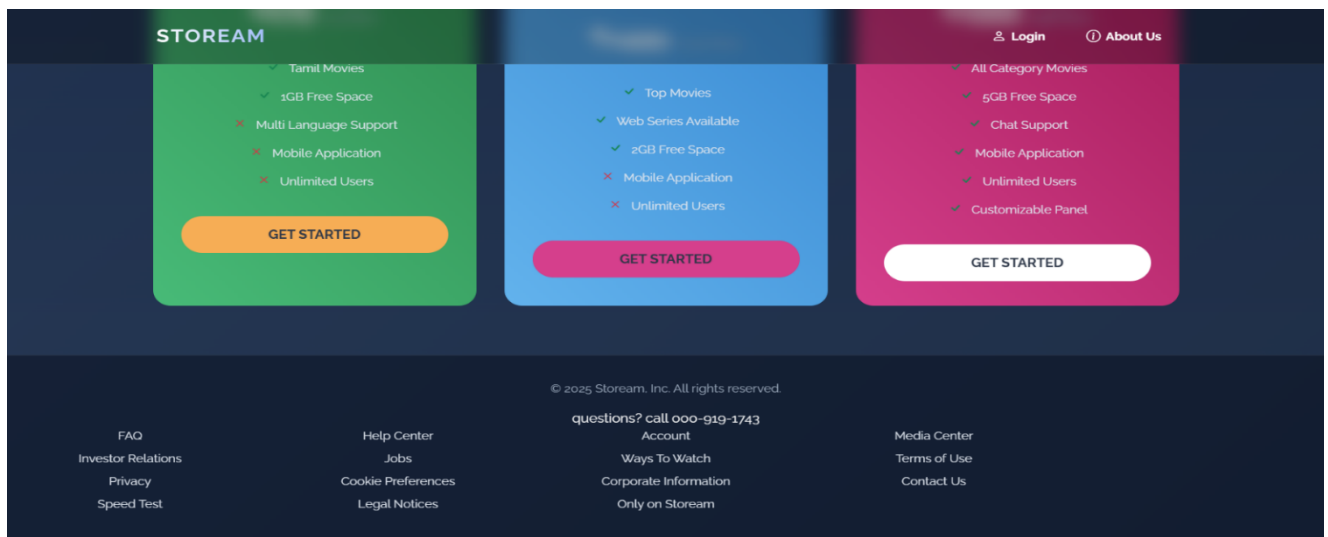
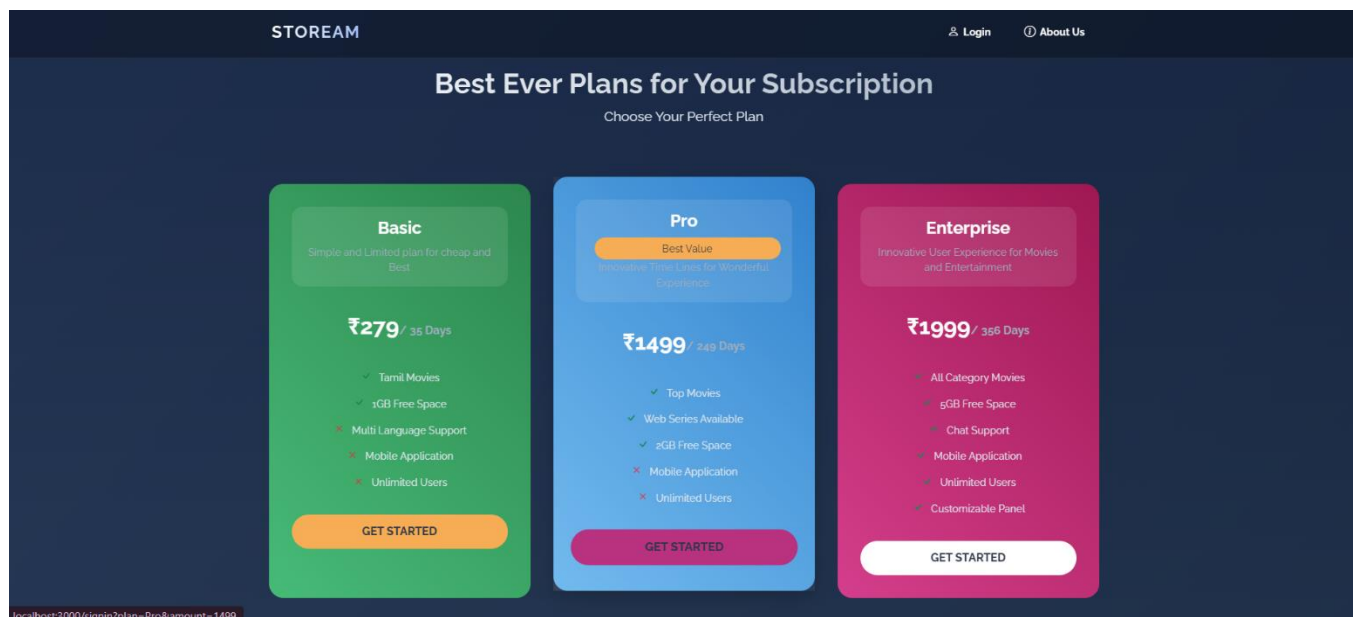
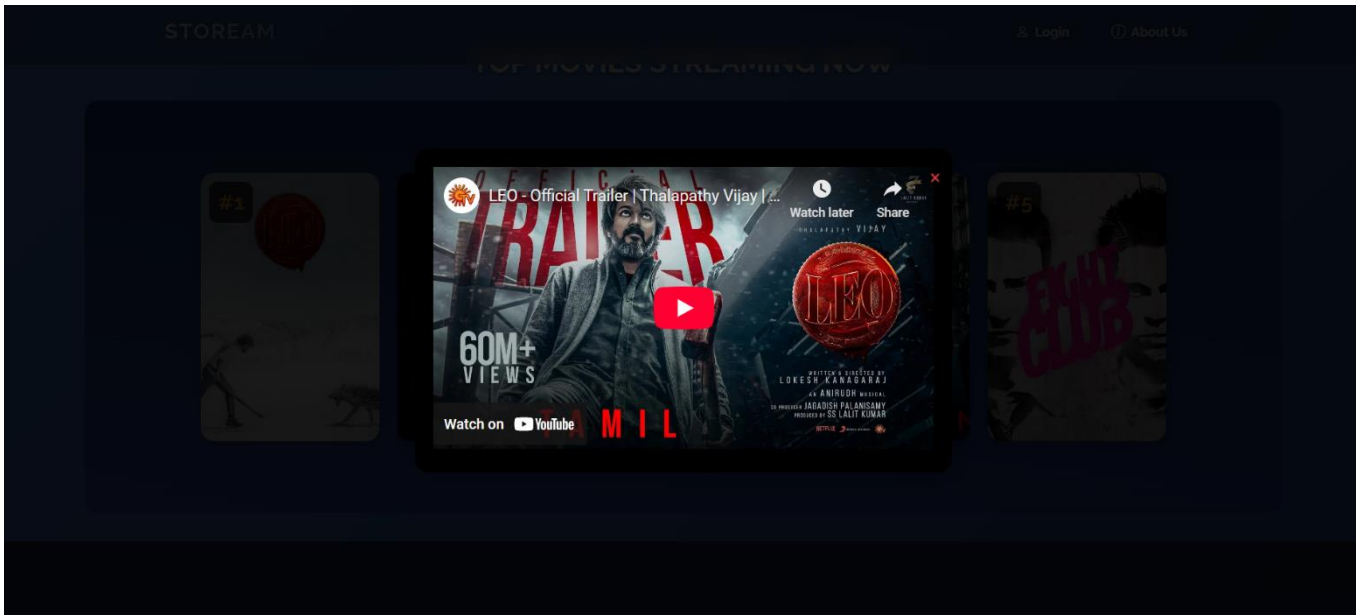
3.3.2 Table Relationship Diagram



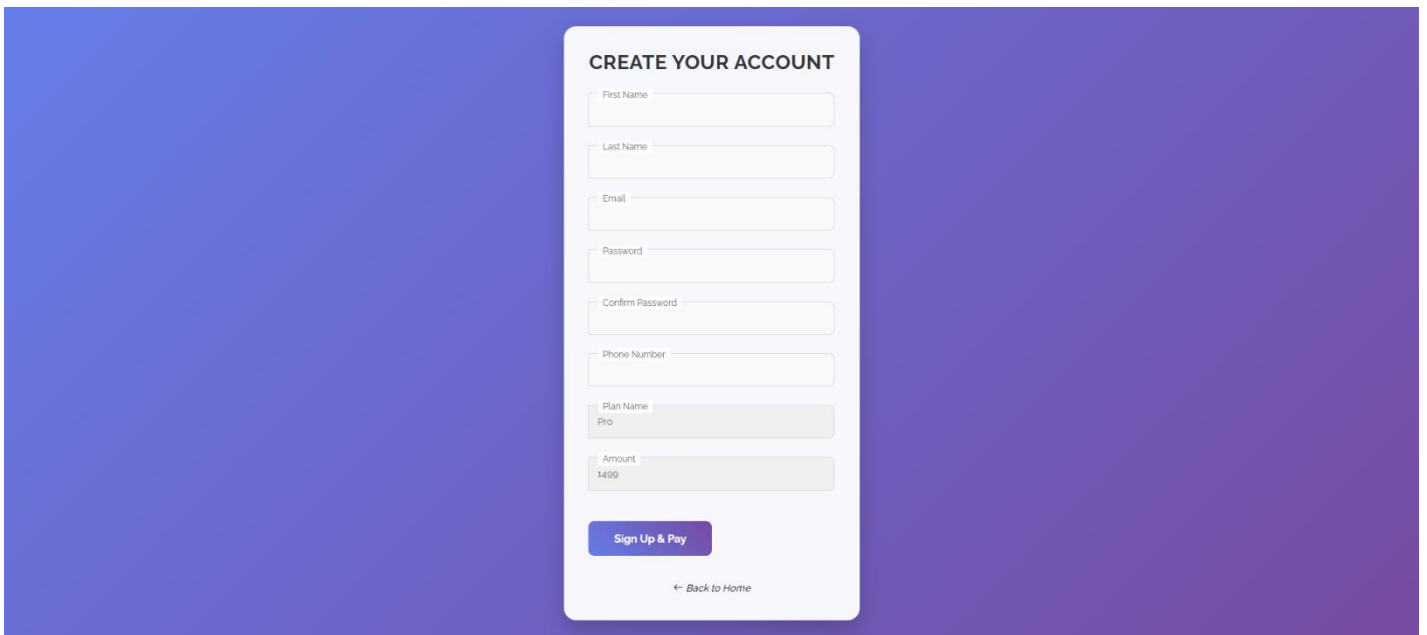
3.4 UI Design

Home Page





Signin Page



CREATE YOUR ACCOUNT

First Name

Last Name

Email

Password

Confirm Password

Phone Number

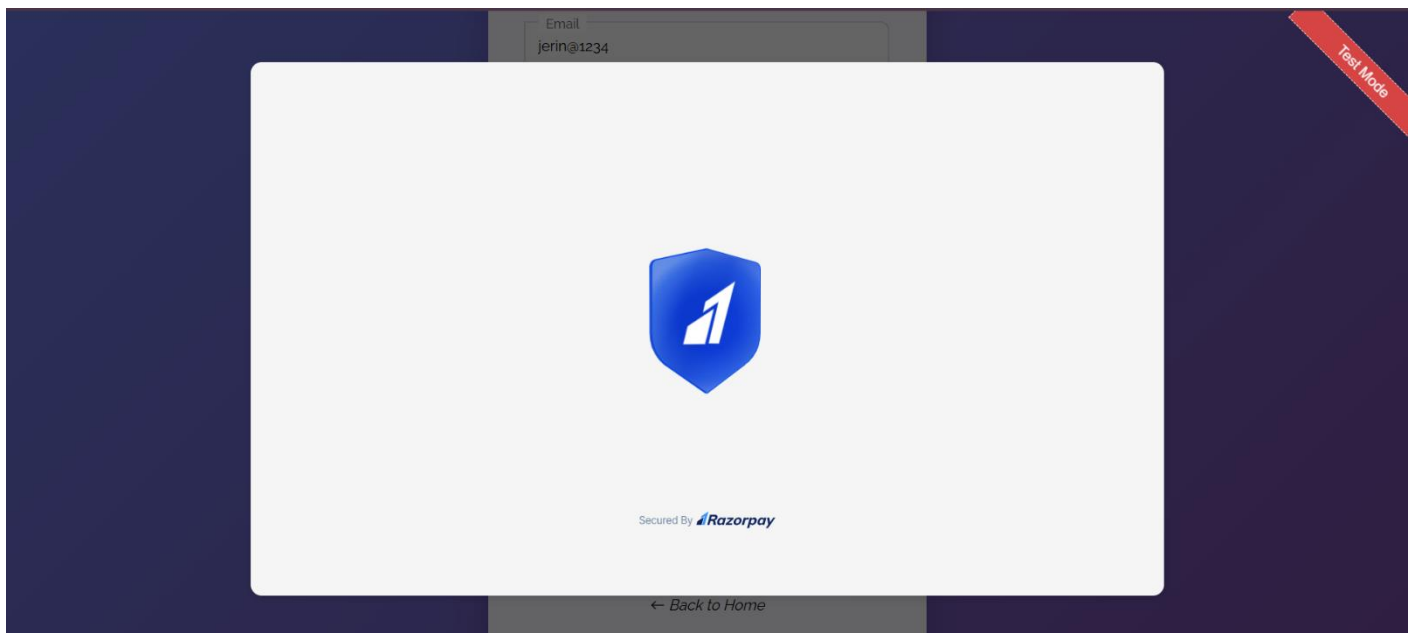
Plan Name
Pro

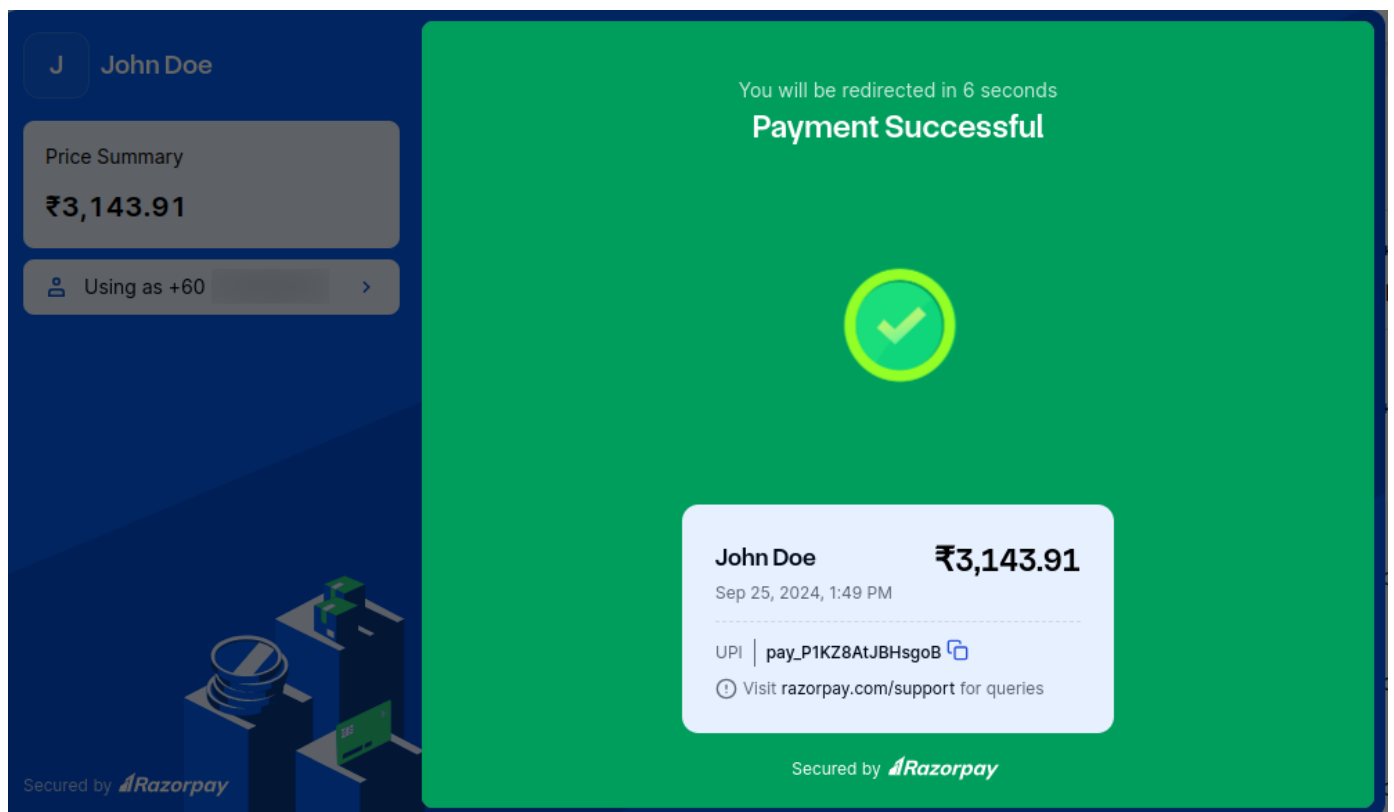
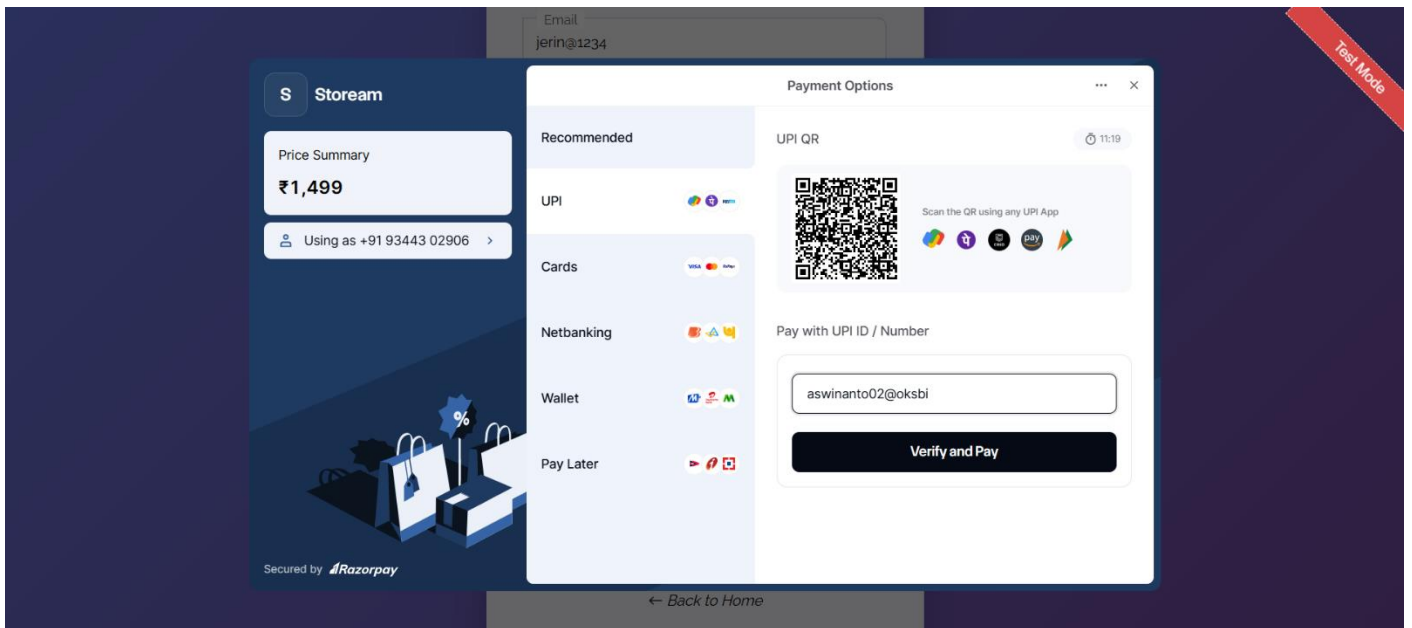
Amount
1499

[Sign Up & Pay](#)

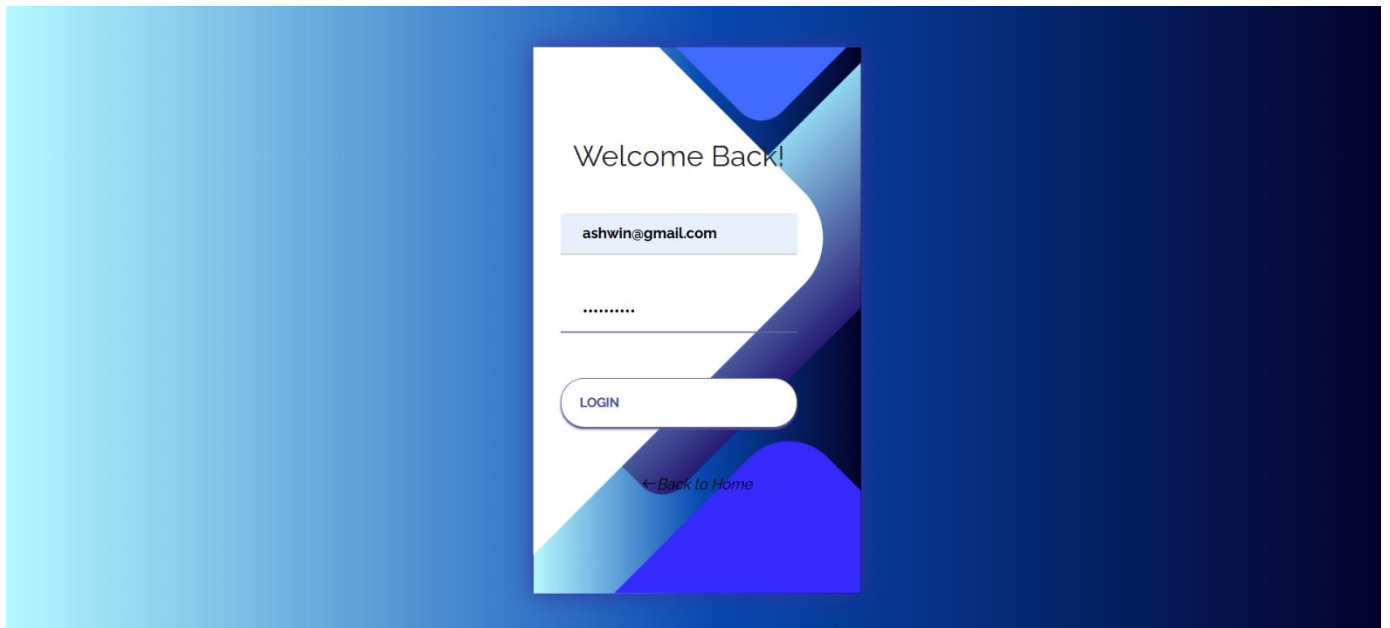
[← Back to Home](#)

Payment Gateway

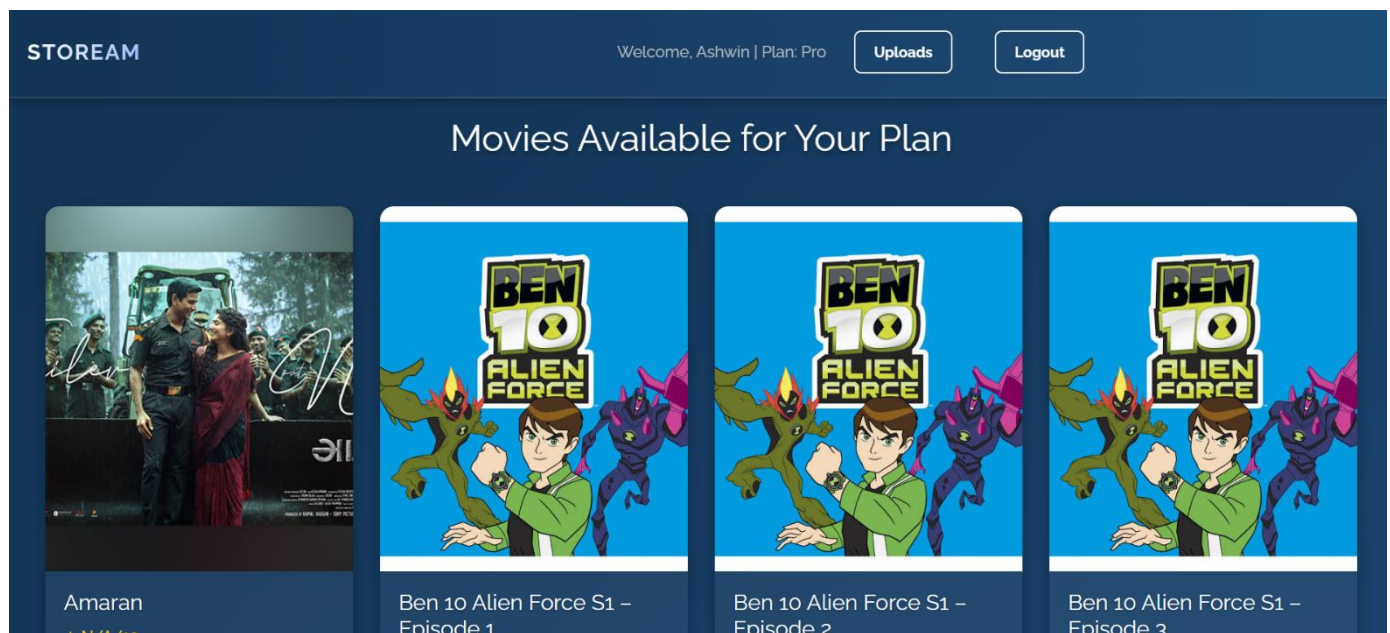




Login Page



Home Page



Upload Section

STOREAMWelcome, Keerthi | Plan: Enterprise

UploadsLogout

User Uploads

Available Space: 3278.71 MB

Choose FileNo file chosen

Upload

ID	File Name	Size (MB)	Action
17	Velaiyilla_Pattathari_2014.mkv	106.88	Delete
18	Thani_Oruvan_2015.mkv	700.12	Delete
19	Oru Naal Koothu (2016).mkv	700.54	Delete
20	Untitled.mp4	333.76	Delete

Movies Available for Your Plan

STOREAMWelcome, sample | Plan: Basic

UploadsLogout

User Uploads

Available Space: 1024.00 MB

Choose FileNo file chosen

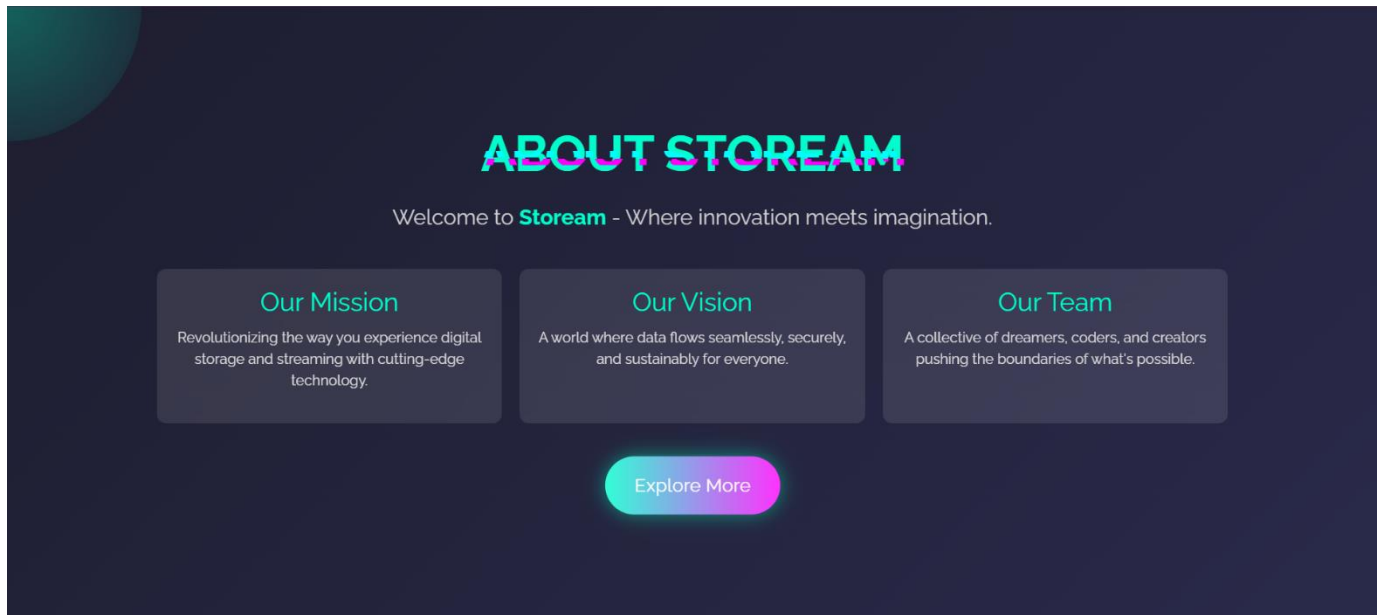
Upload

No uploaded movies yet.

Movies Available for Your Plan



About Page



Admin Page

Add Movie							
ID	Title	Description	Category	Poster	Movie	Release Date	Actions
1	Beauty and Beast	Romantic Movie	Hollywood Movies		View Movie	3/2/2025	Edit Delete
2	Amaran	Army Base	Tamil Movies		View Movie	19/9/2024	Edit Delete
3	Ben 10 Alien Force S1 – Episode 1	Ben 10 Alien Force S1 – Episode 1 – Ben 10 Returns	Web Series		View Movie	2/6/2020	Edit Delete
4	Ben 10 Alien Force S1 – Episode 2	Ben 10 Alien Force S1 – Episode 2 – Ben 10 Returns, Part 2	Web Series		View Movie	15/6/2020	Edit Delete
5	Ben 10 Alien Force S1 – Episode 3	Ben 10 Alien Force S1 – Episode 3 – Max Out	Web Series		View Movie	21/9/2020	Edit Delete

Admin Movie Upload

Description	
	Romantic Movie
	Army Base
Episode 1	Ben 10 Alien Force S1 – Episode 1 – Ben 10 Returns
Episode 2	Ben 10 Alien Force S1 – Episode 2 – Ben 10 Returns
Episode 3	Ben 10 Alien Force S1 – Episode 3 – Max Out
	Tangled - Romantic - Drama - Animation

Add Movie

Title

Description

Category

Tamil Movies

Release Date

dd-mm-yyyy

Poster Image

Choose File | No file chosen

Movie File

Choose File | No file chosen

Add Movie

Category	Poster	Movie	Release Date
Hollywood Movies		View Movie	3/2/2025
Tamil Movies		View Movie	19/9/2024
Web Series		View Movie	2/6/2020
Web Series		View Movie	15/6/2020
Web Series		View Movie	21/9/2020
Animation Movies		View Movie	24/11/2010

Admin Movie edit

Description	
	Romantic Movie
	Army Base
ode 1	Ben 10 Alien Force S1 – Episode 1 – Ben 10 Returns
ode 2	Ben 10 Alien Force S1 – Episode 2 – Ben 10 Returns
ode 3	Ben 10 Alien Force S1 – Episode 3 – Max Out
	Tangled - Romantic - Drama - Animation

Edit Movie

Title

Beauty and Beast

Description

Romantic Movie

Category

Hollywood Movies

Release Date

03-02-2025

Poster Image

Choose File | No file chosen

Movie File

Choose File | No file chosen

Update Movie

Category	Poster	Movie	Release Date	Actions
Hollywood Movies		View Movie	3/2/2025	Edit Delete
Tamil Movies		View Movie	19/9/2024	Edit Delete
Web Series		View Movie	2/6/2020	Edit Delete
Web Series		View Movie	15/6/2020	Edit Delete
Web Series		View Movie	21/9/2020	Edit Delete
Animation Movies		View Movie	24/11/2010	Edit Delete

CHAPTER IV

4.1 Implementation of ReactJs Code

IndexComponent.js

```
import React, { useState } from "react";
import { BrowserRouter as Router, Routes, Route, Navigate } from "react-router-dom";
import HomePage from "../HomeComponent/HomePage";
import LoginComponent from "../LoginComponent/LoginIndex";
import SigninComponent from "../SigninComponent/SigninIndex";
import PaymentPage from "../PaymentComponent/PagementPage";
import MainPage from "../MainComponent/MainPage";
import MainComponent from "../MainComponent/MainIndex";
import AdminComponent from "../AdminComponent/AdminIndex";
import AboutComponent from "../AboutComponent/AboutIndex";
const IndexComponent = () => {
  return (
    <Router>
      <Routes>
        <Route path="/" element={<HomePage />} />
        <Route path="/login" element={<LoginComponent />} />
        <Route path="/signin" element={<SigninComponent />} />
        <Route path="/payment" element={<PaymentPage />} />
        <Route path="/main" element={<MainComponent />} />
        <Route path="/admin" element={<AdminComponent />} />
        <Route path="/about" element={<AboutComponent />} />
      </Routes>
    </Router>
  );
};
export default IndexComponent;
```

HomePage.js

```
import React, { useState } from "react";

import { Link } from "react-router-dom";

import "../HomePage.css"; // For SCSS styling

import "bootstrap/dist/css/bootstrap.min.css";

import "../images/movhead.jpg";

import rounding from "../images/homeround.svg";

export default function HomePage() {

  const [trailerUrl, setTrailerUrl] = useState(null); // State for trailer URL

  const [showTrailer, setShowTrailer] = useState(false); // State for modal visibility

  const plans = [

    { name: "Basic", amount: 279 },

    { name: "Pro", amount: 1499 },

    { name: "Enterprise", amount: 1999 },

  ];

  const topMovies = [

    {

      id: 1,

      title: "Leo",

      image: topmov1,

      description: "A gripping thriller by Lokesh Kanagaraj.",

      genre: "Action/Thriller",

      rating: "9.7",

      trailerUrl: "https://www.youtube.com/embed/Po3jStA673E",
```

```
},  
  
{  
  
  id: 2,  
  
  title: "The Godfather",  
  
  image: topmov2,  
  
  description: "An epic tale of a mafia dynasty.",  
  
  genre: "Crime/Drama",  
  
  rating: "9.2",  
  
  trailerUrl: "https://www.youtube.com/embed/sY1S34973zA",  
  
},  
  
{  
  
  id: 3,  
  
  title: "The Dark Knight",  
  
  image: topmov3,  
  
  description: "A superhero film redefining the genre.",  
  
  genre: "Action/Crime",  
  
  rating: "9.0",  
  
  trailerUrl: "https://www.youtube.com/embed/EXeTwQWrcwY",  
  
},  
  
{  
  
  id: 4,  
  
  title: "Inception",  
  
  image: topmov4,  
  
  description: "A sci-fi thriller that bends the mind.",
```

```

    genre: "Sci-fi/Thriller",
    rating: "8.8",
    trailerUrl: "https://www.youtube.com/embed/8hP9D6kZseM",
  },
  {
    id: 5,
    title: "Fight Club",
    image: topmov5,
    description: "A bold take on consumerism and identity.",
    genre: "Drama",
    rating: "8.8",
    trailerUrl: "https://www.youtube.com/embed/SUXWAEX2jlg",
  },
];

const handleWatchTrailer = (url) => {
  setTrailerUrl(url);
  setShowTrailer(true);
};

const closeTrailer = () => {
  setShowTrailer(false);
  setTrailerUrl(null);
};

function searchMovie() {
  const movieName = document.getElementById('movieSearchInput').value.trim();

```

```

if (!movieName) {

    alert('Please enter a movie name. ');

    return;

}

    fetch(`http://localhost/stoream_api/search_movie.php?movie=${encodeURIComponent(
movieName)}`)

    .then(response => response.json())

    .then(data => {

        if (data.error) {

            alert(data.error);

            return;

        }

        displayMoviePopup(data);

    })

    .catch(error => {

        console.error('Error fetching movie:', error);

        alert('An error occurred while searching for the movie. ');

    });

}

function displayMoviePopup(data) {

    const popup = document.getElementById('moviePopup');

    const movieDetails = document.getElementById('movieDetails');

    const relatedMovies = document.getElementById('relatedMovies');

    const movieCategories = document.getElementById('movieCategories');

```

```

document.getElementById('movieTitle').textContent = data.movie.title || 'Movie Not
Found';

movieDetails.innerHTML = `

<p><strong>Release Date:</strong> ${data.movie.release_date || 'N/A'}</p>

<p><strong>Description:</strong> ${data.movie.description || 'No description
available'}</p>

${data.movie.poster ? `` :
""} `;

relatedMovies.innerHTML = "";

if (data.relatedMovies && data.relatedMovies.length > 0) {

data.relatedMovies.forEach(movie => {

relatedMovies.innerHTML += `<div className="related-movie"
onClick="searchMovieDirect('${movie.title}')">${movie.title}</div>`;

});

} else {

relatedMovies.innerHTML = '<p>No related movies found.</p>';

}

movieCategories.innerHTML = "";

if (data.categories && data.categories.length > 0) {

data.categories.forEach(category => {

movieCategories.innerHTML += `<div className="category-
item">${category}</div>`;

});

} else {

movieCategories.innerHTML = '<p>No categories found.</p>';

}

```

```

        popup.style.display = 'flex';
    }

    function closePopup() {

        document.getElementById('moviePopup').style.display = 'none';

    }

    function searchMovieDirect(movieName) {

        document.getElementById('movieSearchInput').value = movieName;

        searchMovie();

    }

    return (

<div className="hp-homepage">

// Design page

</div>

    );

}

index.php

<?php

header("Access-Control-Allow-Origin: *"); // Allow requests from any origin

header("Access-Control-Allow-Methods: GET, POST, OPTIONS, DELETE, PUT"); //
Allowed methods

header("Access-Control-Allow-Headers: Content-Type, Authorization"); // Allowed headers


include "config.php";

```

```

$method = $_SERVER["REQUEST_METHOD"];

$data = json_decode(file_get_contents("php://input"), true);

if ($method == "GET") {

    // Fetch all movies

    $query = "SELECT * FROM movies";

    $result = mysqli_query($conn, $query);

    if ($result) {

        echo json_encode(mysqli_fetch_all($result, MYSQLI_ASSOC));

    } else {

        echo json_encode(["error" => "Failed to fetch movies"]);

    }

}

if ($method == "POST") {

    // Insert new movie

    if (!empty($data["title"]) && !empty($data["genre"]) && !empty($data["year"])) {

        $title = mysqli_real_escape_string($conn, $data["title"]);

        $genre = mysqli_real_escape_string($conn, $data["genre"]);

        $year = mysqli_real_escape_string($conn, $data["year"]);

        $query = "INSERT INTO movies (title, genre, year) VALUES ('$title', '$genre', '$year')";

        if (mysqli_query($conn, $query)) {

```



```

        echo json_encode(["message" => "Movie added successfully"]);

    } else {

        echo json_encode(["error" => "Failed to add movie"]);

    }

} else {

    echo json_encode(["error" => "Missing required fields"]);

}

}

if ($method == "PUT") {

    // Update movie

    if (!empty($data["id"]) && !empty($data["title"]) && !empty($data["genre"]) &&
!empty($data["year"])) {

        $id = mysqli_real_escape_string($conn, $data["id"]);

        $title = mysqli_real_escape_string($conn, $data["title"]);

        $genre = mysqli_real_escape_string($conn, $data["genre"]);

        $year = mysqli_real_escape_string($conn, $data["year"]);

        $query = "UPDATE movies SET title='$title', genre='$genre', year='$year' WHERE
id='$id'";

        if (mysqli_query($conn, $query)) {

            echo json_encode(["message" => "Movie updated successfully"]);

        } else {

            echo json_encode(["error" => "Failed to update movie"]);

        }

    }

}

```

```

    } else {

        echo json_encode(["error" => "Missing required fields"]);

    }

}

if ($method == "DELETE") {

    // Delete movie

    if (!empty($data["id"])) {

        $id = mysqli_real_escape_string($conn, $data["id"]);

        $query = "DELETE FROM movies WHERE id='$id'";

        if (mysqli_query($conn, $query)) {

            echo json_encode(["message" => "Movie deleted successfully"]);

        } else {

            echo json_encode(["error" => "Failed to delete movie"]);

        }

    } else {

        echo json_encode(["error" => "Missing movie ID"]);

    }

}

mysqli_close($conn);

?>

```

CHAPTER V

5.1 Test Plan

Objective: Stoream OTT platform-la plans, payment, content filtering, admin CRUD, and user experience fully tested.

Scope: Basic, Pro, Enterprise plans, Razorpay integration, admin movie management, user access.

Approach: Manual testing with real scenarios (login, payment, content viewing, admin operations).

Tools: Browser (Chrome/Firefox), Razorpay sandbox, admin dashboard.

Section	Description	Priority
User Authentication	Login, registration, access control	High
Plans & Content	Plan selection, content filtering	High
Payment Gateway	Razorpay payment and space allocation	High
Admin Management	CRUD operations for movies	Medium
Security	Unauthorized access, payment security	High

5.2 Test Cases

1. User Authentication Tests

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-001	User Login - Valid Credentials	1. Open login page 2. Enter valid email/password 3. Click Login	Redirect to dashboard	Pass
TC-002	User Login - Invalid Credentials	1. Open login page 2. Enter wrong email/password 3. Click Login	Error: "Invalid email or password"	Pass
TC-003	User Registration - Valid Details	1. Open registration page 2. Enter valid details 3. Submit	Account created, redirect to login	Pass
TC-004	User Registration - Email Exists	1. Register with existing email	Error: "Email already registered"	Pass

2. Plans & Content Tests

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-005	Basic Plan Content	1. Login 2. Select Basic plan 3. View content	Only Tamil movies display	Pass
TC-006	Pro Plan Content	1. Login 2. Select Pro plan 3. View content	Tamil movies + web series display	Pass

TC-007	Enterprise Plan Content	1. Login 2. Select Enterprise plan 3. View content	Tamil movies, web series, animation, Hollywood display	Pass
TC-008	Plan Selection UI	1. Go to plans page	₹279 (Basic), ₹1499 (Pro), ₹1999 (Enterprise) show	Pass

3. Payment Gateway Tests

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-009	Basic Plan Payment	1. Select Basic plan 2. Pay ₹279 via Razorpay 3. Complete payment	1 GB space allocated, success message	Pass
TC-010	Pro Plan Payment	1. Select Pro plan 2. Pay ₹1499 via Razorpay 3. Complete payment	2 GB space allocated, success message	Pass
TC-011	Enterprise Plan Payment	1. Select Enterprise plan 2. Pay ₹1999 via Razorpay 3. Complete payment	5 GB space allocated, success message	Pass
TC-012	Payment Failure	1. Select plan 2. Enter wrong card details	Error: "Payment failed"	Pass

4. Admin Management Tests

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-013	Add Movie	1. Login as admin 2. Go to dashboard 3. Add movie (e.g., Tamil)	Movie added to category	Pass
TC-014	Edit Movie	1. Select movie 2. Update details	Movie details updated	Pass

TC-015	Delete Movie	1. Select movie 2. Delete	Movie removed from list	Pass
TC-016	User Upload Attempt	1. Login as user 2. Try uploading movie	No upload option visible	Pass

5. Security Tests

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-017	Unauthorized Access	1. Try accessing dashboard without login	Redirect to login page	Pass
TC-018	Admin Access as User	1. Login as user 2. Try accessing admin page	Access denied or redirect	Pass

5.3 Test Report

1. User Authentication I/O Report

Feature	Input	Expected Output
User Login	Email, Password	Dashboard if valid, Error if invalid
User Registration	Name, Email, Password	Account created or "Email exists" error

2. Plans & Content I/O Report

Feature	Input	Expected Output
Basic Plan	Plan selection	Tamil movies only
Pro Plan	Plan selection	Tamil movies + web series
Enterprise Plan	Plan selection	Tamil movies, web series, animation, Hollywood

3. Payment Gateway I/O Report

Feature	Input	Expected Output
Basic Payment	₹279 via Razorpay	1 GB space + success message
Pro Payment	₹1499 via Razorpay	2 GB space + success message
Enterprise Payment	₹1999 via Razorpay	5 GB space + success message
Failed Payment	Wrong card details	"Payment failed" error

4. Admin Management I/O Report

Feature	Input	Expected Output
Add Movie	Movie details (title, category)	Movie added to category
Edit Movie	Updated details	Movie updated
Delete Movie	Delete action	Movie removed
User Upload	User tries uploading	No option available

5. Security I/O Report

Feature	Input	Expected Output
Unauthorized Access	Access without login	Redirect to login
Admin Access	User tries admin page	Access denied

CHAPTER VI

6.1 Implementation and Deployment

Objective: To implement and deploy the Stoream OTT platform on a live server, making it accessible to users.

Scope: Covers ReactJS frontend, PHP backend, Razorpay integration, database setup, and admin controls deployment.

Environment: Production server (e.g., AWS, DigitalOcean, or any hosting service).

6.1.1 Implementation Plan

Step	Task Description	Details
Code Preparation	Prepare Frontend Build	Run npm run build in the ReactJS project to generate the build/ folder.
	Prepare Backend Files	Organize PHP files, configuration, and .htaccess in the backend folder.
Database Setup	Create MySQL Database	Set up tables for movies, users, plans, and payment details (e.g., categories like Tamil, web series).
	Import Initial Data	Upload initial movie categories (Tamil, web series, animation, Hollywood) via admin interface.
Razorpay	Configure Razorpay Live Keys	Switch from sandbox to live mode, update API keys in the backend.
Admin Access	Deploy Admin Dashboard	Ensure admin login and CRUD operations are functional after deployment.

6.1.2 Deployment Steps

Step	Task Description	Actions
Server Setup	Choose Hosting Server	Select a hosting provider like AWS, DigitalOcean, or Hostinger with PHP support.
	Set Up Domain Name	Purchase and configure a domain (e.g., stoream.com) with DNS settings.
	Install SSL Certificate	Enable HTTPS using Let's Encrypt or the hosting provider's SSL option.
File Upload	Upload Frontend Files	Move the React build/ folder to the server root (e.g., /var/www/html/).
	Upload Backend Files	Upload PHP files to a separate folder (e.g., /var/www/backend/).
Server Config	Configure Apache/Nginx	Set up .htaccess or Nginx rewrite rules to remove index.php from URLs.
	Verify PHP Version	Ensure PHP 7.4 or 8.0 is installed with required extensions (e.g., MySQL).
Database Connect	Connect Database	Update database credentials (host, user, password) in the PHP backend.
Testing	Perform Live Testing	Verify login, plan selection, payment, and content filtering work correctly.

CHAPTER VII

7. Conclusion and References

7.1 I/O Report

The Input/Output (I/O) Report summarizes the key features of the Stoream OTT platform, detailing the inputs provided by users and the expected outputs from the system. This ensures that all functionalities align with the project requirements.

1. User Authentication I/O Report

Feature	Input	Expected Output
User Login	Email, Password	Redirect to dashboard if valid, Error message "Invalid email or password" if invalid
User Registration	Name, Email, Password	Account created with success message, or Error "Email already registered" if email exists

2. Plans & Content I/O Report

Feature	Input	Expected Output
Basic Plan Selection	User selects Basic plan	Displays only Tamil movies
Pro Plan Selection	User selects Pro plan	Displays Tamil movies and web series
Enterprise Plan Selection	User selects Enterprise plan	Displays Tamil movies, web series, animation movies, and Hollywood movies (no Hindi)

3. Payment Gateway I/O Report

Feature	Input	Expected Output
Basic Plan Payment	₹279 via Razorpay	1 GB space allocated, Success message
Pro Plan Payment	₹1499 via Razorpay	2 GB space allocated, Success message
Enterprise Plan Payment	₹1999 via Razorpay	5 GB space allocated, Success message
Failed Payment	Invalid card details	Error message "Payment failed"

4. Admin Management I/O Report

Feature	Input	Expected Output
Add Movie	Movie details (title, category)	Movie added to specified category (e.g., Tamil, Hollywood)
Edit Movie	Updated movie details	Movie details updated in the system
Delete Movie	Delete action on movie	Movie removed from the system
User Upload Attempt	User tries uploading movie	No upload option available to users

5. Security I/O Report

Feature	Input	Expected Output
Unauthorized Access	Access dashboard without login	Redirect to login page
Admin Access by User	User tries admin dashboard	Access denied or redirect to user page

7.2 Conclusion

The Stoream OTT platform has been successfully designed, developed, and tested to meet the needs of users seeking an online streaming and storage solution. The system offers three distinct plans—Basic, Pro, and Enterprise—catering to different content preferences and storage requirements. Key features include:

- **Content Accessibility:** Basic plan provides Tamil movies, Pro adds web series, and Enterprise includes animation and Hollywood movies, excluding Hindi content.
- **Payment Integration:** Razorpay ensures secure and seamless transactions, allocating storage (1 GB, 2 GB, 5 GB) based on the selected plan.
- **Admin Control:** The admin has full CRUD capabilities for movie management, ensuring content is curated without user uploads.
- **User Experience:** The ReactJS frontend and PHP backend work together to deliver a responsive and functional interface.

The testing phase confirmed that all functionalities work as intended, with no major issues identified. Deployment on a live server ensures scalability and accessibility. Stoream addresses the problem of limited, plan-based OTT access while providing a secure and efficient platform. Future enhancements could include additional language support, user personalization, or mobile app development.

7.3 References

The following resources were utilized during the development and documentation of the Stoream OTT platform:

1. **ReactJS Documentation** - Official documentation for building the frontend interface.
(<https://reactjs.org/docs/getting-started.html>)
2. **PHP Manual** - Reference for backend development and API integration.
(<https://www.php.net/manual/en/>)
3. **Razorpay Integration Guide** - Guidelines for implementing payment gateway.
(<https://razorpay.com/docs/>)
4. **MySQL Documentation** - Database design and management reference.
(<https://dev.mysql.com/doc/>)
5. **Apache HTTP Server Documentation** - Server configuration and deployment guide.
(<https://httpd.apache.org/docs/>)